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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023
END SEMESTER EXAMINATION - ODD 2023
CS101 - Programming for Problem Solving

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. The Hexadecimal equivalent of the number (101101010010)₂ is (1) CO5 BL4
 - a) b) B52
 - b) A53
 - c) A62
 - d) None of the above

2. Find out the output: (1) CO5 BL4

```
int main()
{
    int i=1;
    printf("\n%d%d%d", i , i++ , ++i);
}
```

 - a) 323
 - b) 331
 - c) 113
 - d) 314

3. Number of bytes required to store a char array of 10 elements is (1) CO2 BL1
 - a) 1 byte
 - b) 10 bytes
 - c) 4 bytes
 - d) 40 bytes

4. What is the output of this C code?

(1) CO5 BL4

```
#include <stdio.h>;  
void main()  
{  
float k=0;  
for(k=0.0; k<3.0; k++);  
printf("%f",k);  
}
```

- a) 3.0
- b) 2.0
- c) 1.0
- d) None of these

5. Members of a Union use

(1) CO4 BL2

- a) same storage locations
- b) different storage locations
- c) no storage locations
- d) different discrete locations

6. Which of the options is correct, if we run the following program:

(1) CO5 BL4

```
#include<stdio.h>;  
void main()  
{  
int num[3]={10,20,30}, *ptr;  
ptr = &num[0]  
printf("%d",*(ptr+3));  
}
```

- a) The output is : 30
- b) The output is: 20
- c) Compiler time error is generated
- d) No output

7. The smallest integer that can be represented using 16-bits in signed 2's complement system is

(1) CO1 BL1

- a) -65536
- b) -32768
- c) -32767
- d) 0

8. What will be the output of the following code? (1) CO2 BL1
- ```
#include<stdio.h>

int main(){

 char ch=321;

 printf("%d",ch);

 return 0 ;

}
```
- a) 65  
b) 66  
c) 321  
d) compilation error
9. How is a string represented in C? (1) CO3 BL3
- a) Integer array  
b) Character array  
c) Floating-point array  
d) Double array
10. (1217)<sub>8</sub> is equivalent to a) b) c) d) (1) CO1 BL1
- a) (1217)<sub>16</sub>  
b) (028F)<sub>16</sub>  
c) (2297)<sub>10</sub>  
d) (0B17)<sub>16</sub>
11. What is the purpose of a function prototype in C? (1) CO4 BL2
- a) To declare the function's return type  
b) To provide the function's implementation  
c) To declare the function's parameters  
d) To declare the function's name
12. The function that is invoked is known as: a) b) c) Called function d) (1) CO2 BL1
- None of these  
a) Calling function  
b) Caller function  
c) Called function  
d) none of them

### Group B

#### (Short Answer Type Questions)

(Answer any three of the following) 3x5=15

13. Define the difference between for loop and do-while loop using example. (5) CO3 BL2
14. Explain with a suitable example on how Call by Reference works (5) CO4 BL3
15. Describe the functions of different units of a digital computer. (5) CO1 BL2

16. What are the roles of the preprocessor in C? (5) CO2 BL2
- 17a. If the starting address of a 2D integer array `a[10][10]` is 2000, what will be the address of `a[7][5]` in row major order representation? (2) CO2 BL3
- 17b. What is Compiler? (2) CO1 BL1
- 17c. Size of character data type is (1) CO3 BL1
- a) 2 bytes
- b) 8 bits
- c) 4 bytes
- d) 8 bytes

**Group C**  
**(Long Answer Type Questions)**  
**(Answer any three of the following) 3x15=45**

18. Write answers to all the questions (15)
- a) Convert the following: Hexadecimal: 3DA to Octal (2) CO1 BL4
- b) Convert the following: Octal: 602 to Binary (2) CO1 BL4
- c) Convert the following: Binary: 10111.1101 to Decimal (4) CO1 BL4
- d) Subtract 1101 -1011 using 2'S Complement. (2) CO1 BL4
- e) For computing the maximum among three numbers, compare between the ternary operator and if-else solution approach (5) CO5 BL4
19. Write answers to all the questions (15)
- a) Write a C program to generate the following pattern. (5) CO5 BL4
- ```

1 2 3 4 5 6 7
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```
- b) Discuss, why goto statement is not frequently used in C programming (5) CO3 BL2
- c) How strings in C is different from character array? (5) CO1 BL1
20. Write Answers for all the questions (15)
- a) Differentiate between structure and union. (3) CO2 BL2
- b) What is type casting. Explain with an example (3) CO2 BL2
- c) Create a structure to specify data on students given below: (9) CO5 BL4
- Roll Number, Name, Department, Course, and Year of admission.
- Assume that there are not more than 450 students in college. Write a program to print names of all students who joined in a particular year.

- | | | | |
|---|------|-----|-----|
| 21a. Write a Program in C to check weather a user given number is even or odd using a user given function. Print will be done in main function. | (7) | CO3 | BL4 |
| 21b. What is recursion? How does it differ from iteration? | (5) | CO3 | BL1 |
| 21c. State the disadvantages of an array | (3) | CO4 | BL1 |
| 22a. Give examples to display the use of strlen(), strcpy(), strcat(), strcmp() and strstr(). | (10) | CO4 | BL4 |
| 22b. Write a program in C to reverse a string using array. | (5) | CO4 | BL1 |
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